Land Cover Change from 1984 to 2024 in the Eastern Panhandle of West Virginia

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Introduction

Understanding what the land cover is important to understand factors for environmental management and urban planning (Thwal, 2019). The best way to visualize these differences are through land cover classification and change detection based on remote sensing images (Thwal, 2019). While these changes could be seen broadly through changes in zoning and development plans, this process is extremely time consuming, especially when different jurisdictions and definitions of zoning requirements come into play (Yasin et al., 2021). These complications have led to the popularity of remote sensing to study urban change, especially with its ability to be periodic, consistent, and most importantly, cost-effective (Saini, 2021).

Urban growth, sadly, is unanimous with environmental change, like clearing forests to become suburban sprawl or even farm land (Yasin et al., 2021; Stuart, 2024). These types of changes can be studied further with satellite images and through the different data collected in these images (Yasin et al., 2021). The best way to see these differences are through spatial temporal analysis (Yasin et al., 2021). Spatial temporal analysis is done in remote sensing by collecting at least two images over the same seasonal time of year for two different years (Yasin et al., 2021). Once these images are collected, they go through a classification process using machine learning algorithms that help group pixels on the image into distinct land cover types and comparing the two images with one another (Thwal, 2019). This type of analysis has been performed for many different areas around the world to help them understand the patterns of land use change that have occurred (Ogashaware & da Silva Brum Bastos, 2012).

Most studies that do spatial-temporal analysis also rely on different composite images and ratios derived from the images collected (Ogashaware & da Silva Brum Bastos, 2012). Composite images are a combination of three bands derived from remote sensing data. The main types of composite images used in spatial temporal analysis are true color, near infrared (NIR) false color, and shortwave infrared (SWIR) false color images (Ogashaware & da Silva

Brum Bastos, 2012). The combination of bands needed to create each composite image can be seen in TABLE 1. Each of the different composites allow for different characteristics of the lands surface to be highlighted. NIR false color images help highlight differences in vegetations types, while SWIR false color images allow for greater distinctions to be seen in urban environments (Ogashaware & da Silva Brum Bastos, 2012). Apart from composite images, ratios can also be used to help highlight differences that might not be seen through composite images. Ratios are derived from performing math on different bands from the image collected (Ogashaware & da Silva Brum Bastos, 2012). In spatial temporal analysis most studies used either normalized difference vegetation index (NDVI), or normalized different built-up index (NDBI), or both (Ogashaware & da Silva Brum Bastos, 2012; Thwal, 2019; Yasin et al., 2020; Yasin et al., 2021). NDVI helps highlight the differences in vegetation types (Ogashaware & da Silva Brum Bastos, 2012; Thwal, 2019). NDBI, on the other hand, helps highlight the difference seen in urban areas (Ogashaware & da Silva Brum Bastos, 2012; Thwal, 2019; Yasin et al., 2020; Yasin et al., 2021). FIGURE 1 shows the formulas needed for NDVI and NDBI. Using both composite images and ratios allow for the algorithms used for the classification process of the spatial temporal analysis to help distinguish between different land cover types that might be visually similar (Ogashaware & da Silva Brum Bastos, 2012).

The land cover classification process preformed in these studies can be broken down into three different types. Supervised classification allows for the algorithm to use training data, collected by the user, to help it in determining what land cover type each pixel should be assigned to (Ogashaware & da Silva Brum Bastos, 2012; Yasin et al., 2020). Unsupervised classification gives the algorithm no training data, rather relies solely on the band values in each pixel to group pixels into groups based on those values (Thwal, 2019; Yasin et al., 2021). After the pixels are grouped the user needs to go group by group to assign a land cover type to each grouping of pixels (Thwal, 2019; Yasin et al., 2021). Lastly, hybrid- supervised classification is a

combination of both methods allowing for mistakes in one process to not over shadow the classification of the image as a whole (Yasin et al., 2021). Regardless of which classification process is done a confusion matrix is performed after each classification to ensure the accuracy of the algorithm and be able to proceed with further analysis (Thwal, 2019; Stuart, 2024).

Materials and Methods

In the state of West Viriginia there are only three counties that are seeing tremendous growth: Berkeley, Jefferson, and Morgan (Culeyhouse & Lockhart, 2023). These three counties are collectively known as the Eastern Panhandle of West Virigina (Culeyhouse & Lockhart, 2023). Over the past few decades, they have seen growth not seen anywhere else in the state (Culeyhouse & Lockhart, 2023). This growth is attributed to being in close proximity to Washington, D.C., while still having nature be a major attraction in the area (Culeyhouse & Lockhart, 2023). The growth comes with the cost-of-living skyrocketing throughout the Eastern Panhandle (Culeyhouse & Lockhart, 2023). Raising costs are causing many to describe the new developments as “pushing people out, not bringing them in” (Culeyhouse & Lockhart, 2023). Understanding where this growth is occurring through remote sensing could be the key in managing the growth of this region (Culeyhouse & Lockhart, 2023; Yasin et al., 2021).

This study collected two images to understand where growth has occurred through a spatial temporal analysis. Parameters for image selection needed to be established before moving forward. These images needed to have less than 5% of cloud cover for them to be used in land cover classifications (Yasin et al., 2020; Yasin et al., 2021). Apart from cloud cover, the season needed to be selected as well as the years for the analysis. Spring and summer images are considered to be better given that vegetation is in full bloom, allowing for better distinctions between land cover types (Ogashaware & da Silva Brum Bastos, 2012; Stuart, 2024). Lastly, to show the greatest difference in land cover type change, it was determined to go as far back as

possible using Landsat images, which started in 1984 (U.S.G.S., 2024). The most modern image, likewise, was selected from Landsat images, being 2024 at the time of this study (U.S.G.S., 2024). Placing these parameters into Earth Explorer to view Landsat images it was determined that the area of study was split between two different image frames, tile grid 8 and 9 (U.S.G.S., 2024). Both frames needed to be able to fulfil the established parameters for the given time frame. The images were finally selected with the first being from 1984/09/19 and the second image from 2024/09/09. After downloading the images, the first images were then composited together for all available bands to allow for viewing images in different composites later in the analysis process. Once the composites were created, they were then mosaiced together into a single image. Finally, a clip was preformed over the mosaiced image to only view the counties in the Eastern Panhandle of West Virginia. This process was performed for both years of study. The resulting images can be seen mosaiced, clipped, and composited as true color images on MAP 1.

Once the images were processed, composites could be viewed using the different band combinations, seen in TABLE 1. Apart from composites being viewed, ratios were also created to help in the analysis. NDVI and NDBI, whose formulas can be seen in FIGURE 1, were both created for both years to help highlight differences. NDVI allows for greater sensitivity to chlorophyll in plants, which reflect NIR, and reducing the noise seen in images by normalization the results between -1 and +1 (Ogashaware & da Silva Brum Bastos, 2012; Yasin et al., 2021). Values closer +1 are deemed to be green and healthy vegetation, while values closer to -1 are indicative of unhealthy vegetation or no vegetation (Ogashaware & da Silva Brum Bastos, 2012; Yasin et al., 2021). NDBI on the other hand provides the same benefits of normalization between -1 and +1, but for man made areas that have higher SWIR (Yasin et al., 2020; Yasin et al., 2021). Values closer to +1 show a higher amount of SWIR, indicative of built-up areas, while

values closer to -1 tend to be natural features (Yasin et al., 2020; Yasin et al., 2021). Both NDVI and NDBI ratio images for both years can be seen on MAP 2.

The next part of the analysis was determining which land cover classification type will be used. After reviewing the studies, it was determined that a hybrid-supervised classification method will be used for the most effective means to classification (Ogashaware & da Silva Brum Bastos, 2012; Stuart, 2024). Hybrid-supervised classification combines both supervised classification and unsupervised classification methods (Saini, 2021). The number of original classes were 6 classes, being water, forest, urban, vegetation, farm, and other (Thwal, 2019). Training data was drawn over a SWIR composite for both years, with a minimum of 30 figures of each group being created to help in the classification (Saini, 2021). Once the training data was collected a supervised classification algorithm needed to be selected. “[A] Maximum Likelihood Classification (MLC) algorithm is the most popular and conventional method to perform a land use and land cover classification task with an acceptable rate of accuracy” (Stuart, 2024). This algorithm was used by 3 of the studies referenced, while the remaining three used Support Vector Machine (SVM) (Ogashaware & da Silva Brum Bastos, 2012; Thwal, 2019; Yasin et al., 2020; Yasin et al., 2021; Saini, 2021; Stuart, 2024). The deciding factor was in the fact that SVM algorithm have been shown to get better accuracy when compared to using the MLC algorithm (Stuart, 2024). When creating the supervised classification using the SVM algorithm, an NDVI image was added into the analysis to help in the algorithm’s groupings. Next an unsupervised classification was performed over the same data using an ISODATA clustering algorithm (Saini, 2021; Stuart, 2024). A total of 35 spectral clusters were created, that were grouped based on a NDBI image and band number similarities (Saini, 2021). Both classification methods were then combined into one land cover classification. Classes that did not align were altered based on visual confirmations using image composites. Once the final image classification was performed an accuracy assessment was performed through the use of a confusion matrix, see TABLES 2-5

(Stuart, 2024). A confusion matrix compares what land cover type is occurring at a specific pixel, compared to what the classification process result was (Stuart, 2024). In the image from 1984, 516 pixels were looked at and 518 were looked at for the image from 2024. The specific pixels were selected through an equalized random sampling method, which allows for a proportion of each land cover type compared to the total size to be selected (Saini, 2021). After performing the classifications, it is clear to see in TABLEs 2 and 3, that an over accuracy over 80% was not met. An 80% accuracy was determined to be needed to be able to continue with the analysis (Yasin et al., 2020). Looking at the tables it was clear that the easiest fix would be to combine farm and vegetation classes into one vegetation class. TABLEs 4 and 5 show the results of combining the classes together and MAPs 3 and 4 show the final classifications. Both classes accuracy are greater or equal to 80% accuracy allowing for the last aspect of spatial temporal analysis can be performed. The new image was then subtracted from the old image allowing for the creation of FIGURE 2 showing the land cover changes that have occurred between 1984 and 2024.

Results and Discussion

When it comes to the confusion matrix most studies tended to get overall accuracy results in the 90% to 99% range (Thwal, 2019; Saini, 2021; Stuart, 2024). During this analysis the lowest over accuracy is seen in TABLE 2 and TABLE 4 with 72% and 69% respectively. Studying these matrix's closer it can be seen in TABLE 2 that the greatest omission and commission error between two classes was between farm and vegetation. These errors indicate that these classes similarities are causing the classification to have difficulty to be able to differentiate between the two classes. After combining the farm and vegetation class in the 1984 classification a new confusion matrix was created, see TABLE 3. This new 1984 confusion matrix shows that the overall accuracy grew to 87%. Now looking at the original 2024 confusion matrix, TABLE 4, the greatest omission and commission error between two classes was close to

a tie between 2 combinations. Vegetation and forest had the greatest number of omission and commission errors, but vegetation and farm weren't too far behind them. Stuart (2024) had their greatest commission and omission error between vegetation and forest, showing that this commonly happens. But, because the 1984 classification needed to group vegetation and farm, the same was done for the 2024 classification. TABLE 5 shows the final confusion matrix of 2024 showing the overall accuracy of 80%, just barely high enough to continue the study.

The land cover change seen from 1984 to 2024 is not surprising given the amount of growth that has occurred in the region over this 40-year span (Culeyhouse & Lockhart, 2023). FIGURE 2 shows the change in classifications between 1984 to 2024. TABLE 6 explains the change better, by showing the percent of land cover represented by each type for the year. This table shows that while roughly 3,400 acres of forest and almost 5,000 acres of vegetation have changed classification during these 40 years, they only lost 1% each of their total area coverage. The urban classification on the other hand grained 2% of land cover or 9,700 acres, showing that the growth the area is having is causing urban land cover to expand (Culeyhouse & Lockhart, 2023). Other studies tended to break up the study periods in roughly 10- or 5-year intervals rather and a large 40-year jump (Saini, 2021; Stuart, 2024). Regardless, they too trended to show that growing populations results in a loss of natural vegetation and a rise in urban area (Yasin et al., 2021; Saini, 2021; Stuart, 2024).

When using spatial temporal analysis to address land cover change for a region the results need to be taken with a grain of salt. "Land cover classification using pixel-based approach has limitations such as mixed pixels and salt and pepper effects" (Thwal, 2019). Mixed pixels are pixels, like many seen in this classification process, that due to its resolution of 30 m by 30 m, a pixel can contain a mixture of features on the ground. One common doubt had was trying to be consistent in deciding where the border between a forest and farmland was. Another problem that occurred was in urban areas, where a street might be tree lined causing confusion in what

those pixels needed to be classified as. This confusion in classifications over border areas and in areas that have other factors like tree lined streets can create a salt and pepper effect (Thwal, 2019). This effect is seen with the classification of one pixel being surrounded by other classes, like forest surrounded by farm land. The best way to resolve all these problems would be through having image resolution that is much smaller than the 30 m by 30 m used for this analysis (Stuart, 2024). Sharper resolution isn't always available, especially looking at older images, so these problems need to be understood.

Conclusions

The opinion that developers are making homes by removing cornfields and orchards could be seen as more plausible seeing that the loss in forest and vegetation roughly equals the amount of growth seen by urban areas (Culeyhouse & Lockhart, 2023). Using this information can allow the region to reflect if they are willing to continue to expand out or whether the region wants to look into growing up (Yasin et al., 2021). Cities that grow up don't show a loss of vegetation, but still continue to see population to grow through taller buildings (Yasin et al., 2021).

Performing another analysis over this same area should be done with breaking up the 40-year timeframe to see where the growth is occurring most closely during each decade at least. This should also be done using only 3 classes, rather than the 7 classes that were originally used in this study, to be able to really focus on seeing where the land cover changes are occurring (Saini, 2021). Lastly, confusion matrix should also be performed after each classification method to ensure that SVM for supervised classification and ISODATA for unsupervised classification are the best methods. This also helps to make sure that the training data for the supervised classification is adequate and whether the correct number of clusters were created for the unsupervised classification.

References

Ogashaware, I. & da Silva Brum Bastos, V. (2012) A Quantitative Approach for Analyzing the Relationship between Urban Heat Islands and Land Cover. *Remote Sensing*, *4*, 3596-3618. doi:10.3390/ rs4113596

Thwal, N.S., Ishikawa, T., & Watanabe, H. (2019) Land Cover Classification and Change Detection Analysis of Multispectral Satellite Images Using Machine Learning. *Image and Signal Processing for Remote Sensing*, *XXV*, edited by Bruxxon, L. Production of SPIE. Vol. 11155, 111551O.

Yasin, M. Y., Abdullah, J., Noor, N. M., & Yusoff, M. M. (2020) Land Cover and NDBI analysis to map built up area in Iskandar Malaysia. *IOP Conf. Series: Earth and Environmental Science*, *540*. doi:10.1088/1755-1315/540/1/012073

Yasin, M.Y., Abdullah, J., Noor, N. M., Yusoff, M. M., & Noor N. M. (2021) Landsat observation of urban growth and land use change using NDVI and NDBI analysis. *IOP Conf. Series: Earth and Environmental Science*, *1067*.

Saini, V. (2021) Mapping Environmental Impacts of Rapid Urbanisation and Deriving Relationship between NDVI, NDBI and Surface Temperature: A Case Study. JESSD 2021. *IOP Conf. Series: Earth and Environmental Science*, *940*.

Stuart, W., Hossain, A. K. M. A., Hunt, N., Mix, C. & Qin, H. (2024) Spatiotemporal Analysis of Urban Forest in Chattanooga, Tennessee from 1984 to 2021 Using Landsat Satellite Imagery. *Remote Sens*, *16*, *2419*. https://doi.org/10.3390/ rs16132419

Culeyhouse, H. & Lockhart, P. R. (2023) West Virginia's Eastern Panhandle is full of opportunity – and struggling with growing pains. The Register~ Heraold. 2023. Oct. 20. https://www.register-herald.com/news/life/west-virginia-s-eastern-panhandle-is-full-of-

opportunity-and-struggling-with-growing-pains/article_67e93f4e-6eb3-11ee-8854-379ad9a7d115.html

United States Census Bureau (2024) TIGER/Line Shapefiles: West Virginia Counties

U.S.G.S. (2024) Earth Explorer

Landsat 9 (2024) 2024-09-09. GRID 8 Vertical & 27 Horizontal

Landsat 9 (2024) 2024-09-09. GRID 9 Vertical & 27 Horizontal

Landsat 5 (1984) 1984-09-19. GRID 8 Vertical & 27 Horizontal

Landsat 5 (1984) 1984-09-19. GRID 9 Vertical & 27 Horizontal

Landsat Missions (2021) Common Landsat Band Combinations. Pub. U.S.G.S. https://www.usgs.gov/media/images/common-landsat-band-combinations

U.S.G.S. (2024) What are the band designations for the Landsat satellites? Pub. U.S.G.S. https://www.usgs.gov/faqs/what-are-band-designations-landsat-satellites

Maps, Tables, and Figures

Table 1. Band Combinations Needed for Different Composites. True color images use red, green and blue bands. NIR false color composites use NIR band as red, red band as green and green band as blue. Lastly, SWIR false color composites use SWIR band as red, NIR band as green, and green band as blue.

Year (Landsat #)	Type of Composite	Red	Green	Blue
1984 (Landsat 5)	True Color Composite	Band 3	Band 2	Band 1
	NIR False Color Composite	Band 4	Band 3	Band 2
	SWIR False Color Composite	Band 5	Band 4	Band 2
2024 (Landsat 9)	True Color Composite	Band 4	Band 3	Band 2
	NIR False Color Composite	Band 5	Band 4	Band 3
	SWIR False Color Composite	Band 6	Band 5	Band 3

Table 2. Original 1984 Confusion Matrix. Shows the omission and commission errors created as the hybrid-supervised classification process. The resulting overall accuracy was only 72%.

	water	forest	urban	farm	vegetation	other	total	U_accuracy
water	10	0	0	0	0	0	10	100%
forest	1	255	3	11	10	0	280	91%
urban	1	0	12	3	1	1	18	67%
farm	0	14	2	43	23	1	83	52%
vegetation	0	2	8	56	48	1	115	42%
other	0	0	7	0	0	3	10	30%
total	12	271	32	113	82	6	516	
P_accuracy	83%	94%	38%	38%	59%	50%		72%

Table 3. Final 1984 Confusion Matrix. Shows the omission and commission errors created as the hybrid-supervised classification process after combining the farm and vegetation classes to vegetation class. The resulting overall accuracy was 87%.

	water	forest	urban	vegetation	other	total	U_accuracy
water	10	0	0	0	0	10	100%
forest	1	255	3	21	0	280	91%
urban	1	0	12	4	1	18	67%
vegetation	0	16	10	170	2	198	86%
other	0	0	7	0	3	10	30%
total	12	271	32	195	6	516	
P_accuracy	83%	94%	38%	87%	50%		87%

Table 4. Original 2024 Confusion Matrix. Shows the omission and commission errors created as the hybrid-supervised classification process. The resulting overall accuracy was only 69%.

	Water	forest	urban	farm	vegetation	other	total	U_accuracy
water	10	0	0	0	0	0	10	100%
forest	2	225	5	11	34	0	277	81%
urban	0	0	22	4	1	1	28	79%
farm	0	5	4	48	19	1	77	62%
vegetation	0	23	11	34	48	0	116	41%
other	4	0	0	0	0	6	10	60%
total	16	253	42	97	102	8	518	
P_accuracy	63%	89%	52%	49%	47%	75%		69%

Table 5. Final 2024 Confusion Matrix. Shows the omission and commission errors created as the hybrid-supervised classification process after combining the farm and vegetation classes to vegetation class. The resulting overall accuracy was 80%.

	Water	forest	urban	vegetation	other	total	U_accuracy
water	10	0	0	0	0	10	100%
forest	2	225	5	45	0	277	81%
urban	0	0	22	5	1	28	79%
vegetation	0	28	15	149	1	193	77%
other	4	0	0	0	6	10	60%
total	16	253	42	199	8	518	
P_accuracy	63%	89%	52%	75%	75%		80%

Table 6. Land Cover Change from 1984 to 2024. The area in acres for each class for each year is shown. A percentage of what those acres represent as a percent of land cover over the whole of the Eastern Panhandle. The Change columns show the changes seen from 1984 to 2024, in acres and as percentages. The last columns show that out of the acres changed what percentage of land cover each class received, like showing that out of all that classes that gained acres the urban class gained 99% of them.

	1984		2024		Change		Acres Changed	
Water	2,105	0%	2,177	0%	72	0%	+1%	
Forest	273,851	56%	270,448	55%	-3403	-1%		-36%
Urban	17,377	4%	27,156	6%	9779	2%	+99%	
Vegetation	193,689	40%	188,699	39%	-4990	-1%		-52%
Other	1,326	0%	140	0%	-1186	0%		-12%

$$\mathrm{NDVI} = \frac{(\mathrm{NIR}-\mathrm{Red})}{(\mathrm{NIR}+\mathrm{Red})} \qquad \mathrm{NDBI} = \frac{(\mathrm{SWIR}-\mathrm{NIR})}{(\mathrm{SWIR}+\mathrm{NIR})}$$

Figure 1. NDVI and NDBI Formulas. Shows the formulas used to create the ratios of NDVI and NDBI.

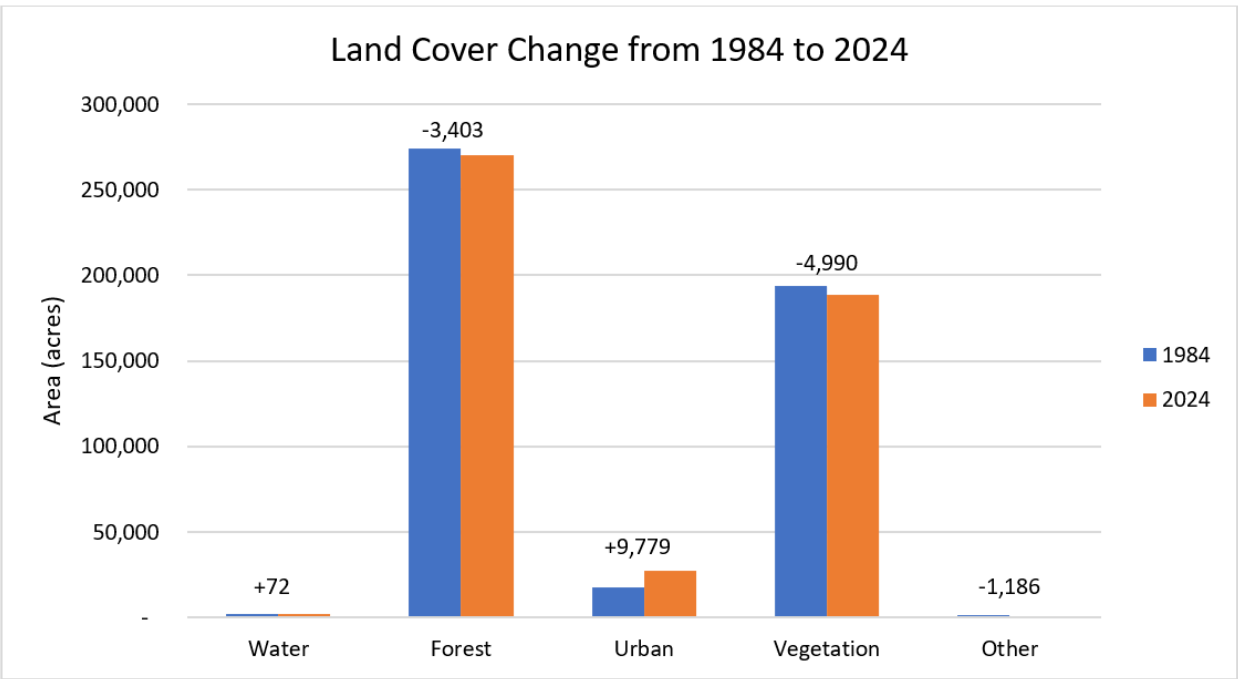
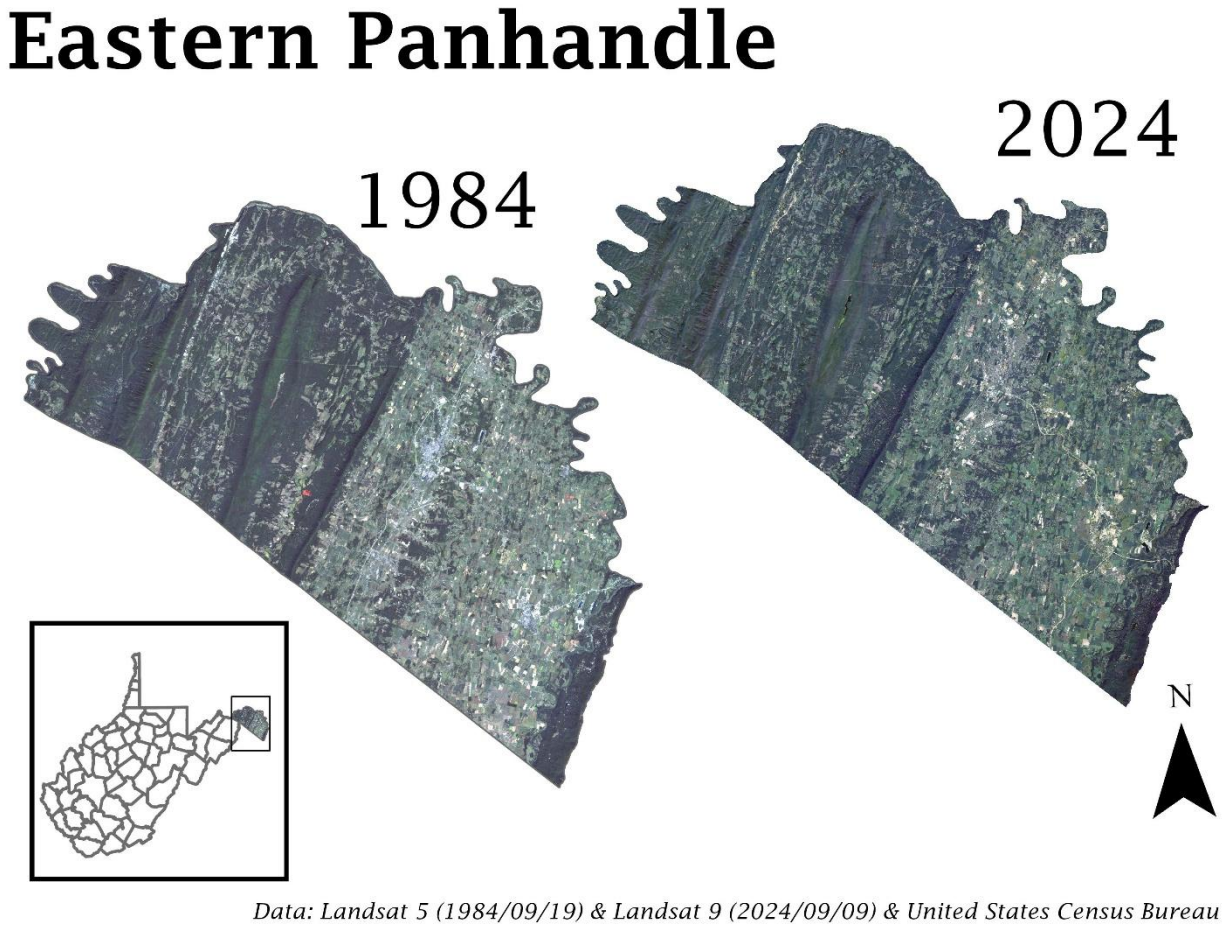
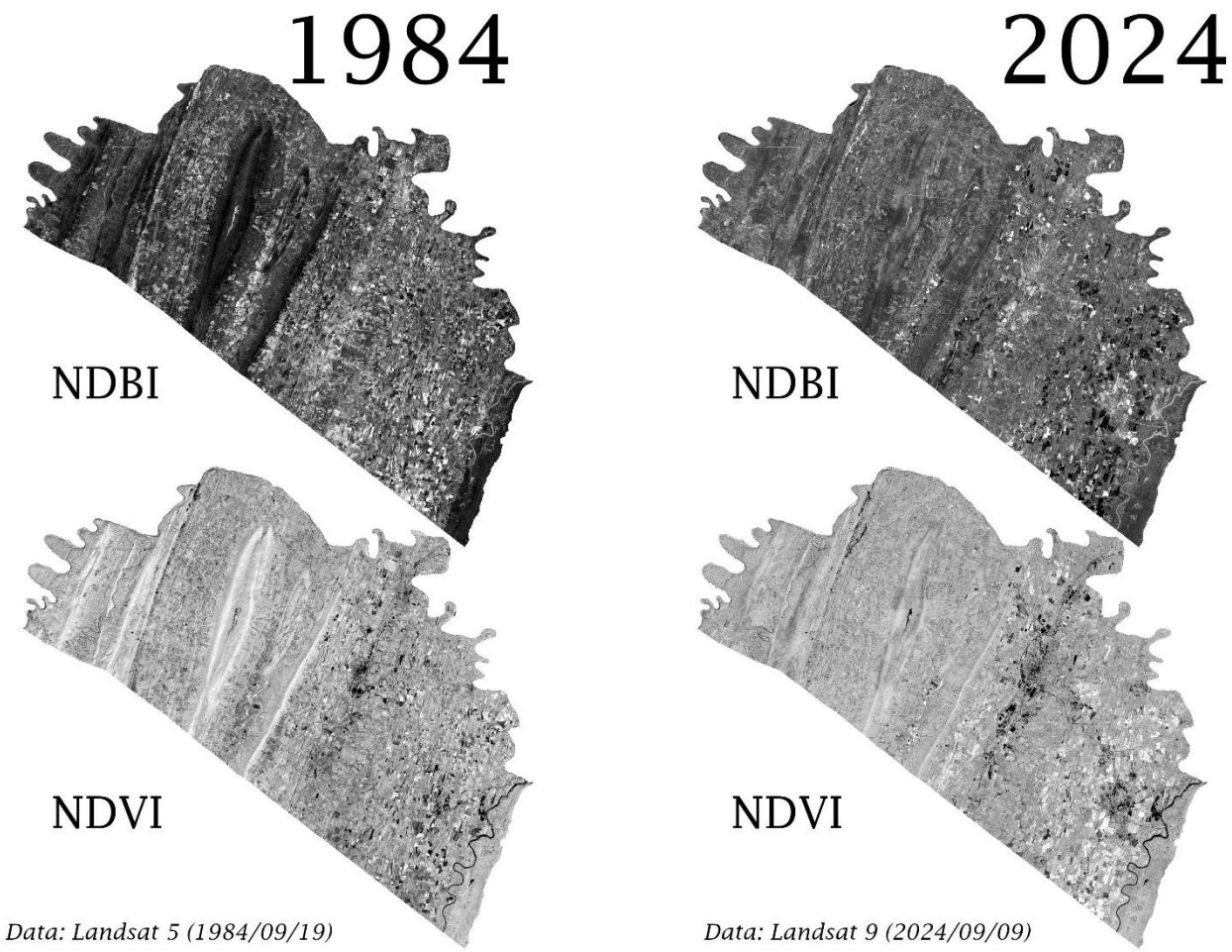


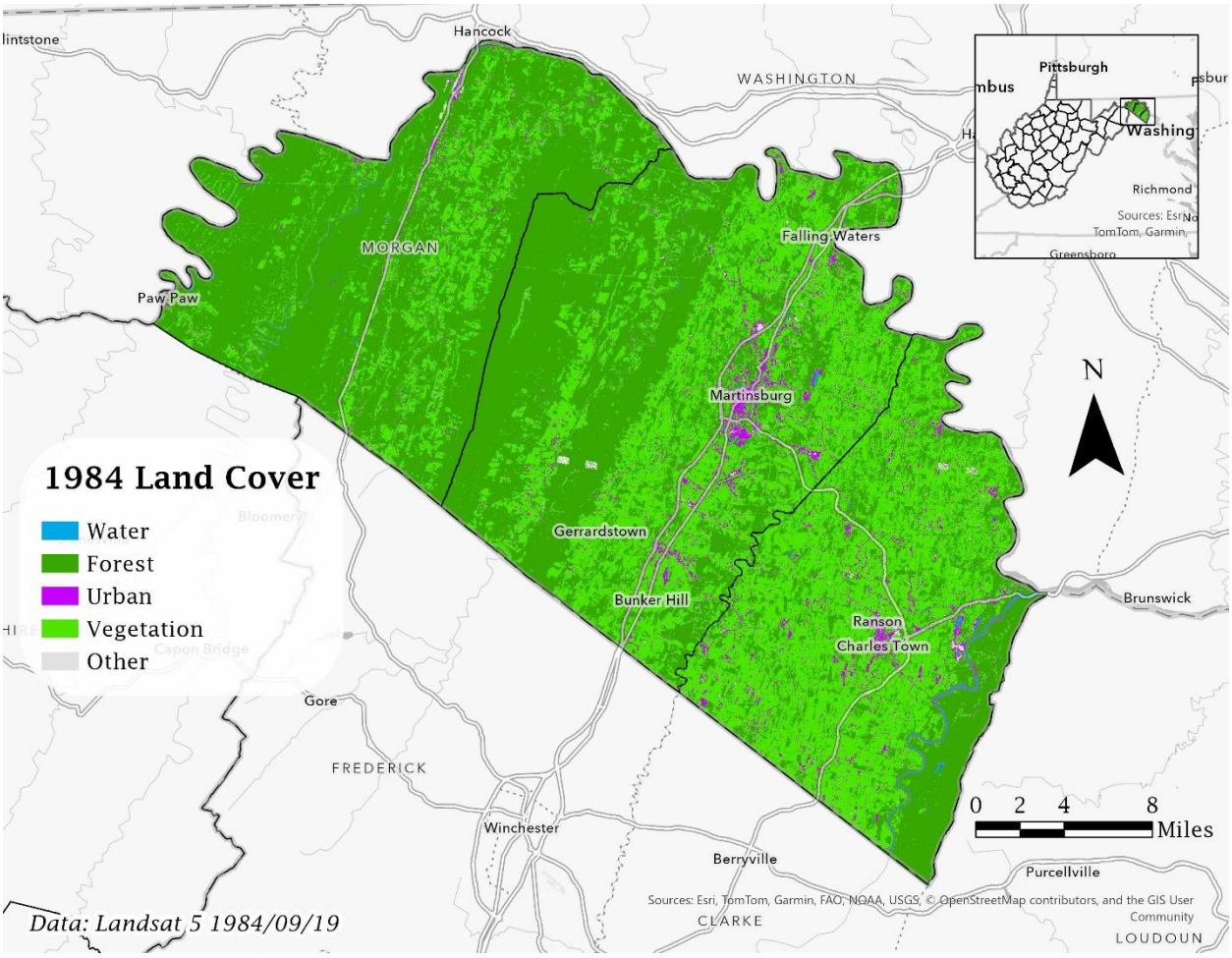
Figure 2. Land Cover Change From 1984 to 2024. Shows the acres for each class divided by year, as well as the number of acres lost or gained between the 40-year span.



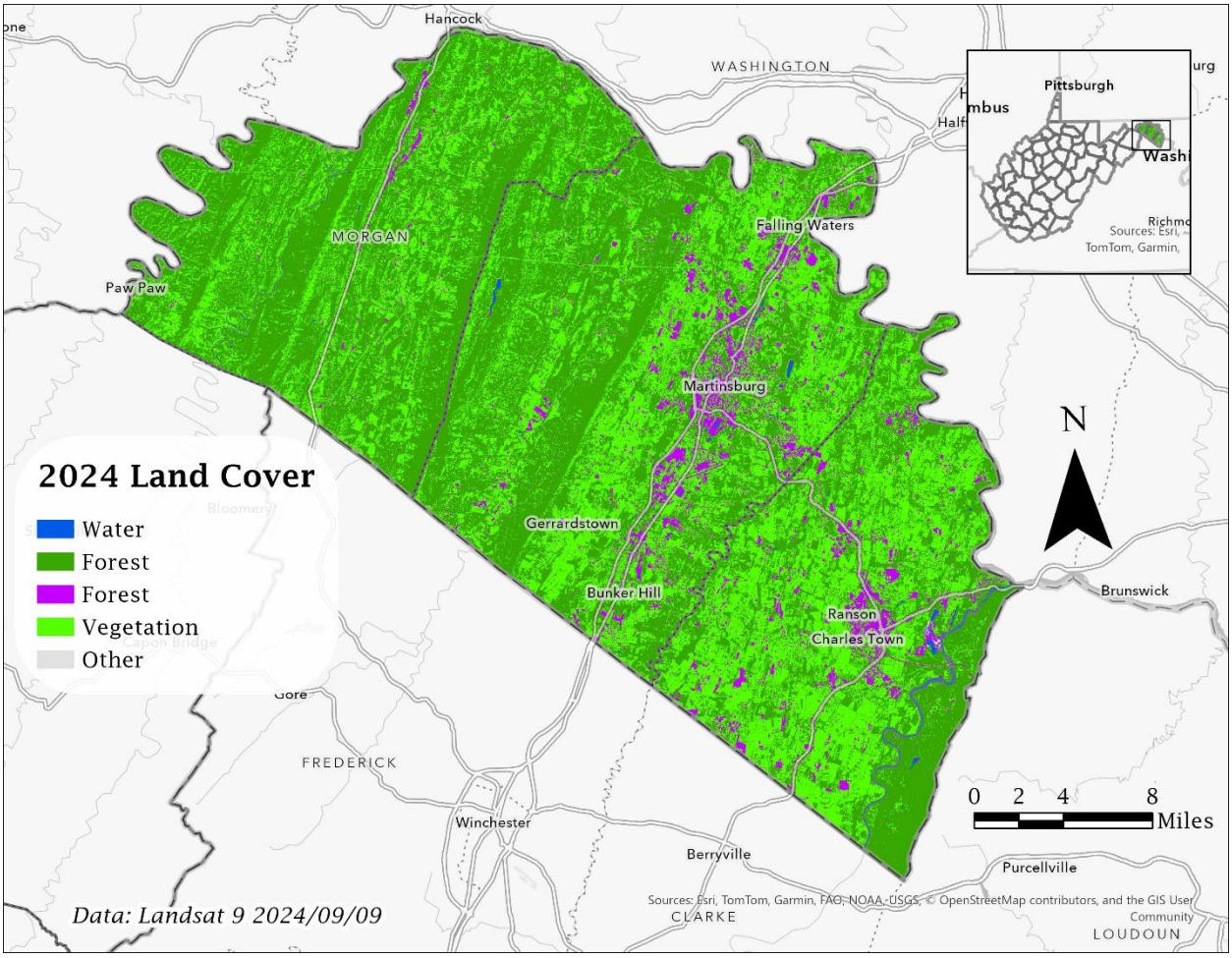
Map 1. Eastern Panhandle. Both 1984 and 2024 true color composites of the mosaiced and clipped images for the study area of the Eastern Panhandle.



Map 2. NDBI and NDVI Images. Both 1984 and 2024 NDBI and NDVI ratio derived images. White areas in NDBI represent urban areas, while white for NDBI represents lush vegetation.



Map 3. Hybrid-Supervised Classification Land Cover for 1984. The final representation of the classification process preformed.



Map 4. Hybrid-Supervised Classification Land Cover for 2024. The final representation of the classification process preformed.